

Top Oscilloscopes Launched in 2014 - Q1 2015

Company	Model	Features	Price
GW Instek	GDS-300/GDS-200 compact digital storage oscilloscopes 	- Equipped with USB storage function, full touchscreen with two analogue signal input channels; advanced and standard models come with 70MHz, 100MHz and 200MHz bandwidths - Maximum sample rate per channel is 1GSa/s and memory depth is 5Mpts - GDS-300 series oscilloscopes are equipped with 50,000 counts DMM and GDS-200 5000 counts DMM	Starts from US\$ 1195
Keysight Technologies	InfiniVision 3000T X-Series 	- Upgradable bandwidths from 100MHz to 1.0GHz, capacitive touchscreen - Comes with six-instruments-in-one integration, including oscilloscope functionality, digital channels (MSO), protocol analysis capability, a digital voltmeter, a WaveGen function/arbitrary waveform generator and an 8-digit hardware counter/totalsiser - Supports a wide range of popular and emerging serial bus applications: MIL-STD 1553 and ARINC 429, I2S, CAN/CAN-FD/CAN-Symbolic, LIN, SENT, FlexRay, RS232/422/485/UART and I2C/SPI	Starts from ₹ 235,037
Pico Technology	PicoScope 9341 	- Ideal for measuring skew and crosstalk between differential pairs on serial buses such as HDMI, SATA and PCIe - Data rates up to 40Gb/s can be viewed, with 5th-harmonic characterisation possible up to 8Gb/s - Scope is controlled through its LAN or USB interface using PicoSample 3 software - Two or four channels, 20GHz (17.5ps) bandwidth, 1MS/s sample rate to 32kS store, 64fs, 15THz effective sample rate	US\$ 22,395.45
Rohde & Schwarz	RTE digital oscilloscopes 	- Bandwidths ranging from 200MHz to 2GHz including two- and four-channel models with 1.5GHz and 2GHz bandwidth - Comes with 5GSa/s sampling rate per channel and up to 200Msample memory 26.4cm (10.4-inch) XGA screen, sensitivity of up to 500µV/div and vertical resolution of 16-bit in high-definition mode	N/A
Tektronix	DPO7000SX 70 GHz ATI performance oscilloscopes 	70GHz real-time oscilloscope with Tektronix' patented asynchronous time interleaving (ATI) technology 200GS/s sample rate with 5ps/sample resolution for improved resolution and timing - Flexibility to use one channel at 70GHz, 200GS/s or two channels at 33GHz, 100GS/s	US\$ 315,000
Tektronix	MDO3000 series 	- Features two or four analogue input channels with bandwidth ranging from 100MHz to 1GHz, 16 digital channels (optional) and one RF input channel matching the bandwidth of the oscilloscope - Offers more than 280,000 waveforms per second capture rate - Integrated digital voltmeter (DVM) provides 4-digit AC RMS, DC or AC+DC RMS voltage and 5-digit frequency measurements - Nine optional serial protocol analysis packages: I2C, SPI, RS-232, USB 2.0, CAN, LIN, FlexRay, MIL-STD-1553 and audio	Starts from ₹ 248,000

In order to further boost productivity, the interface also includes an alphanumeric touchpad that replaces knob based operation and touch based interaction, facilitating higher flexibility in displaying measurement information.

Teddyne LeCroy and GW Instek are among the other vendors to include touchscreen capability in their latest oscilloscopes. In January 2015, GW Instek debuted two knob-free, completely touch based oscilloscopes with 70MHz, 100MHz and 200MHz bandwidths. GDS-200/300 series of

oscilloscopes are highly suitable for professional on-field assignments as well as research and development (R&D) labs.

Graphical triggers. With signals under test getting more convoluted, and digital speeds and device complexity continuing to rise, it becomes demanding for engineers to separate inconsistencies in their devices. Intuitive graphical triggers, earlier not incorporated in mainstream oscilloscopes, aim to help engineers debug and identify the nature of their cutting-edge devices faster and

in a simpler manner. With graphical triggers, an engineer can use a finger to create a trigger by drawing a box around the signal of interest on the oscilloscope's display.

High on integration. The trend slowly seems to be shifting to integration of multiple T&M tools into one instrument.

In the recent past, multiple vendors like Tektronix and Keysight Technologies have introduced such tools. Last year, Tektronix launched their MDO3000 series that comes with six-instruments-in-one box.